UC Berkeley

Department of Electrical Engineering and Computer Sciences

EECS 127 - EECS 227AT. Optimization Models in Engineering

Lecture Note 22

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In general, for the dual feasibility condition $\lambda_i^* \geq 0,\ i = 1, \ldots, m$, there're 2^m possibilities to find which λ_i^*'s are zero and which are strictly positive. This approach is not efficient and there are numerical algorithms to solve the KKT conditions more efficiently.

Example 1. Consider $\mathbf{A} \in \mathbb{R}^{m \times n}$ where $\mathbf{A}$ has full row rank (fat matrix). The constraint $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ is underdetermined if $m < n$. We want to find the minimum norm solution:

$$\min \|\boldsymbol{x}\|_2 \quad \text{s.t. } \mathbf{A}\boldsymbol{x} = \boldsymbol{b}.$$

We used projection before and it was hard. KKT is a shortcut.
Let the rows of $\mathbf{A}$ be $\boldsymbol{a}_1^{\mathrm{T}}, \ldots, \boldsymbol{a}_m^{\mathrm{T}}$. The optimization problem can be written as

$$\begin{aligned} \min \quad & \boldsymbol{x}^{\mathrm{T}}\boldsymbol{x}, \\ \text{s.t.} \quad & \boldsymbol{a}_1^{\mathrm{T}}\boldsymbol{x} = b_1, \\ & \vdots \\ & \boldsymbol{a}_m^{\mathrm{T}}\boldsymbol{x} = b_m. \end{aligned}$$

Let $\mu_1^*, \ldots, \mu_m^*$ be the Lagrange multipliers. The KKT conditions are:

(1) $\mathbf{A}\boldsymbol{x}^* = \boldsymbol{b}$.

(2) doesn't exist since there's no inequalities.

(3) doesn't exist since there's no inequalities.

(4)

$$\begin{aligned} & \underbrace{\nabla(\boldsymbol{x}^{\mathrm{T}}\boldsymbol{x})\big|_{\boldsymbol{x}=\boldsymbol{x}^*}}_{2\boldsymbol{x}^*} + \sum_{j=1}^{m} \mu_j^* \underbrace{\nabla(\boldsymbol{a}_j^{\mathrm{T}}\boldsymbol{x})\big|_{\boldsymbol{x}=\boldsymbol{x}^*}}_{\boldsymbol{a}_j} = \mathbf{0} \\ \Leftrightarrow \quad & 2\boldsymbol{x}^* + \underbrace{\begin{bmatrix} \boldsymbol{a}_1 & \cdots & \boldsymbol{a}_m \end{bmatrix}}_{\mathbf{A}^{\mathrm{T}}} \underbrace{\begin{bmatrix} \mu_1^* \\ \vdots \\ \mu_m^* \end{bmatrix}}_{\boldsymbol{\mu}^*} = \mathbf{0} \\ \Leftrightarrow \quad & \boldsymbol{x}^* = -\frac{1}{2}\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu}^*. \end{aligned}$$

By (1) and (4), we get

$$-\frac{1}{2}\mathbf{A}\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu}^* = \boldsymbol{b} \ \Rightarrow\ \boldsymbol{\mu}^* = -2(\mathbf{A}\mathbf{A}^{\mathrm{T}})^{-1}\boldsymbol{b}.$$

Here $\mathbf{A}\mathbf{A}^{\mathrm{T}}$ is invertible since $\mathbf{A}$ has full row rank. Substituting this into (4), we get

$$\boldsymbol{x}^* = -\frac{1}{2}\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu}^* = \mathbf{A}^{\mathrm{T}}(\mathbf{A}\mathbf{A}^{\mathrm{T}})^{-1}\boldsymbol{b}.$$

This coincides the solution we obtained in Lecture Note 5. ■

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More general case: quadratic minimization with equality constraints:

$$\begin{aligned} \min \quad & \boldsymbol{x}^{\mathrm{T}}\mathbf{P}_0\boldsymbol{x} + \boldsymbol{q}_0^{\mathrm{T}}\boldsymbol{x} + r_0, \\ \text{s.t.} \quad & \mathbf{A}\boldsymbol{x} = \boldsymbol{b}. \end{aligned}$$

- Slater's condition: there exists $\boldsymbol{y}$ such that $\mathbf{A}\boldsymbol{x} = \boldsymbol{b}$ (same as feasibility).
- Convexity of the objective: $\mathbf{P}_0 \succeq \mathbf{0}$.
- KKT conditions:

  (1) $\mathbf{A}\boldsymbol{x}^* = \boldsymbol{b}$.

  (2) doesn't exist since there's no inequalities.

  (3) doesn't exist since there's no inequalities.

  (4)

$$\begin{aligned} &\nabla(\boldsymbol{x}^{\mathrm{T}}\mathbf{P}_0\boldsymbol{x} + \boldsymbol{q}_0^{\mathrm{T}}\boldsymbol{x} + r_0)\big|_{\boldsymbol{x}=\boldsymbol{x}^*} + \sum_{j=1}^{m} \mu_j^* \nabla(\boldsymbol{a}_j^{\mathrm{T}}\boldsymbol{x})\big|_{\boldsymbol{x}=\boldsymbol{x}^*} \\ = 2\mathbf{P}_0\boldsymbol{x}^* + \boldsymbol{q}_0 + &\underbrace{\sum_{j=1}^{m} \boldsymbol{a}_j \mu_j^*}_{\mathbf{A}^{\mathrm{T}}\boldsymbol{\mu}^*} = \mathbf{0}. \end{aligned}$$

  Combining (1) and (4), we get

$$\underbrace{\begin{bmatrix} \mathbf{A} & \mathbf{0}_{m\times m} \\ 2\mathbf{P}_0 & \mathbf{A}^{\mathrm{T}} \end{bmatrix}}_{(n+m)\times(n+m)} \begin{bmatrix} \boldsymbol{x}^* \\ \boldsymbol{\mu}^* \end{bmatrix} = \begin{bmatrix} \boldsymbol{b} \\ -\boldsymbol{q}_0 \end{bmatrix}.$$

  This system of equations could have no solution (e.g., when $\mathbf{P}_0 = \mathbf{0}$, the optimal objective is $-\infty$), 1 solution, or infinitely many solutions. However, KKT is necessary and sufficient so this equation characterizes all solutions.

Optimality conditions for Lasso:

$$\min \ \underbrace{\|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2 + \alpha\|\boldsymbol{x}\|_1}_{f(\boldsymbol{x})}.$$

- The objective function $f(\boldsymbol{x})$ is convex, but we can't use $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$ since $\|\boldsymbol{x}\|_1$ is not differentiable if $x_i = 0$ for some i.
- We can use subdifferential but it is not part of this class.
- Our approach: convert Lasso to QP (using the epigraph technique in Lecture Note 18) and use KKT:

$$\begin{aligned} \text{Lasso} \iff \min_{\boldsymbol{x},\boldsymbol{t}\in\mathbb{R}^n} \quad & \boldsymbol{x}^{\mathrm{T}}\underbrace{\boxed{\mathbf{P}_0}}_{\mathbf{A}^{\mathrm{T}}\mathbf{A}}\boldsymbol{x} + \underbrace{\boxed{\boldsymbol{q}_0^{\mathrm{T}}}}_{-2\boldsymbol{b}^{\mathrm{T}}\mathbf{A}}\boldsymbol{x} + \underbrace{\boxed{r_0}}_{\boldsymbol{b}^{\mathrm{T}}\boldsymbol{b}} + \alpha\sum_{i=1}^{n} t_i, \\ \text{s.t.} \quad & -t_i \le x_i \le t_i, \ i = 1, \ldots, n. \end{aligned}$$

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The constraints can be written as

$$\begin{aligned} x_i - t_i \le 0, \quad i = 1, \dots, n; \\ -t_i - x_i \le 0, \quad i = 1, \dots, n. \end{aligned}$$

Let $\overline{\lambda}_i^*$ and $\underline{\lambda}_i^*$ be the Lagrange multipliers for the first and second constraints, respectively. The KKT conditions are:

(1) $x_i^* - t_i^* \le 0$, $-t_i^* - x_i^* \le 0$, $i = 1, \dots, n$.
(2) $\overline{\lambda}_i^*, \underline{\lambda}_i^* \ge 0$, $i = 1, \dots, n$.
(3) $\overline{\lambda}_i^*(x_i^* - t_i^*) = 0$, $\underline{\lambda}_i^*(-t_i^* - x_i^*) = 0$, $i = 1, \dots, n$.
(4) $\nabla_{\boldsymbol{x},\boldsymbol{t}} L(\boldsymbol{x}, \boldsymbol{t}, \overline{\boldsymbol{\lambda}}^*, \underline{\boldsymbol{\lambda}}^*)\big|_{(\boldsymbol{x},\boldsymbol{t})=(\boldsymbol{x}^*,\boldsymbol{t}^*)} = \mathbf{0}$, where

$$L(\boldsymbol{x}, \boldsymbol{t}, \overline{\boldsymbol{\lambda}}^*, \underline{\boldsymbol{\lambda}}^*) = \boldsymbol{x}^{\mathrm{T}} \mathbf{P}_0 \boldsymbol{x} + \boldsymbol{q}_0^{\mathrm{T}} \boldsymbol{x} + r_0 + \alpha \sum_{i=1}^n t_i + \sum_{i=1}^n \overline{\lambda}_i^*(x_i - t_i) + \sum_{i=1}^n \underline{\lambda}_i^*(-t_i - x_i)$$

and

$$\nabla_{\boldsymbol{x},\boldsymbol{t}} L = \begin{bmatrix} \nabla_{\boldsymbol{x}} L \\ \nabla_{\boldsymbol{t}} L \end{bmatrix} = \begin{bmatrix} \nabla_{\boldsymbol{x}} L \\ \frac{\partial L}{\partial t_1} \\ \vdots \\ \frac{\partial L}{\partial t_n} \end{bmatrix} = \mathbf{0}.$$

Also,

$$\begin{aligned} \nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{t}, \overline{\boldsymbol{\lambda}}^*, \underline{\boldsymbol{\lambda}}^*) &= 2\mathbf{P}_0 \boldsymbol{x} + \boldsymbol{q}_0 + \overline{\boldsymbol{\lambda}}^* - \underline{\boldsymbol{\lambda}}^*, \\ \frac{\partial L(\boldsymbol{x}, \boldsymbol{t}, \overline{\boldsymbol{\lambda}}^*, \underline{\boldsymbol{\lambda}}^*)}{\partial t_i} &= \alpha - \overline{\lambda}_i^* - \underline{\lambda}_i^*, \ i = 1, \dots, n. \end{aligned}$$

So, (4) can be written as

$$2\mathbf{P}_0 \boldsymbol{x}^* + \boldsymbol{q}_0 + \overline{\boldsymbol{\lambda}}^* - \underline{\boldsymbol{\lambda}}^* = \mathbf{0}, \tag{*}$$

$$\alpha - \overline{\lambda}_i^* - \underline{\lambda}_i^* = 0, \ i = 1, \dots, n. \tag{**}$$

- For the constraint $|x_i| \le t_i$, we know that due to epigraph, at optimality we must have $|x_i^*| = t_i^*$ (that's why the reformulation does not change the solution).
- There are three scenarios:

  (i) $x_i^* > 0$, then $-t_i^* - x_i^* = -|x_i^*| - x_i^* < 0$. By (3), we have

  $$-\underline{\lambda}_i^*(-t_i^* - x_i^*) = 0 \ \Rightarrow \ \underline{\lambda}_i^* = 0.$$

  Substituting this into (**), we get

  $$\alpha - \overline{\lambda}_i^* - \underline{\lambda}_i^* = 0 \ \Rightarrow \ \overline{\lambda}_i^* = \alpha.$$

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(ii) $x_i^* < 0$, similarly we can obtain $\overline{\lambda}_i^* = 0$ and $\underline{\lambda}_i^* = \alpha$.

(iii) $x_i^* = 0$. In this case, (1) and (3) automatically hold since

$$t_i^* = |x_i^*| = 0 \;\Rightarrow\; -t_i^* - x_i^* = x_i^* - t_i^* = 0.$$

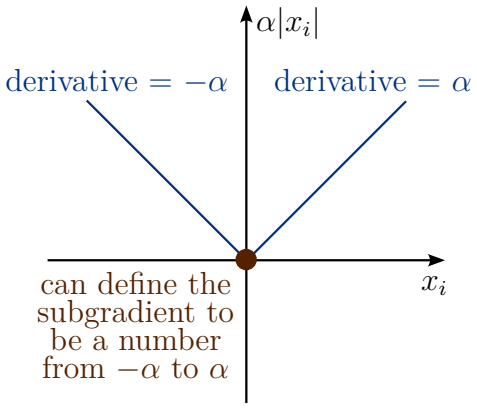
By (2) and (**), we have

$$\alpha - \overline{\lambda}_i^* - \underline{\lambda}_i^* = 0,\ \overline{\lambda}_i^*, \underline{\lambda}_i^* \geq 0 \Rightarrow \overline{\lambda}_i^*, \underline{\lambda}_i^* \in [0, \alpha]$$
$$\Rightarrow \overline{\lambda}_i^* - \underline{\lambda}_i^* \in [-\alpha, \alpha].$$

- In summary: let $\boldsymbol{\lambda}^* := \overline{\boldsymbol{\lambda}}^* - \underline{\boldsymbol{\lambda}}^*$. Due to the above discussion and (*), KKT is equivalent to: $\boldsymbol{x}^*$ is a solution to Lasso iff there exists $\boldsymbol{\lambda}^* = [\lambda_1^*, \ldots, \lambda_n^*]^{\mathrm{T}}$ such that

$$2\mathbf{P}_0\boldsymbol{x}^* + \boldsymbol{q}_0 + \boldsymbol{\lambda}^* = \mathbf{0} \quad \text{and} \quad \lambda_i^* \begin{cases} = \alpha, & \text{if } x_i^* > 0, \\ = -\alpha, & \text{if } x_i^* < 0, \\ \in [-\alpha, \alpha], & \text{if } x_i^* = 0. \end{cases}$$

- Remark: $2\mathbf{P}_0\boldsymbol{x}^* + \boldsymbol{q}_0 + \boldsymbol{\lambda}^*$ in the condition is composed of the gradient of $\|\mathbf{A}\boldsymbol{x} - \boldsymbol{b}\|_2^2$ and subgradient of $\alpha\|\boldsymbol{x}\|_1$. We obtained this directly through KKT.



KKT is a necessary and sufficient condition. Let's prove sufficiency (necessity is difficult).

Proof. Consider

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \leq 0,\ i = 1, \ldots, m, \\ & h_j(\boldsymbol{x}) = 0,\ j = 1, \ldots, k. \end{aligned}$$

Assume a point $\boldsymbol{x}^*$ satisfies the KKT conditions (1)–(4). We prove that $\boldsymbol{x}^*$ is optimal.

- Due to KKT condition (1), $\boldsymbol{x}^*$ is a feasible point.
- Consider an arbitrary feasible point $\boldsymbol{y}$ (i.e., $f_i(\boldsymbol{y}) \leq 0, h_j(\boldsymbol{y}) = 0$). If we show that $f_0(\boldsymbol{x}^*) \leq f_0(\boldsymbol{y})$, it proves that $\boldsymbol{x}^*$ is a global minimum.

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- By KKT condition (4), we have $\nabla_{\boldsymbol{x}} L(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)\big|_{\boldsymbol{x}=\boldsymbol{x}^*} = \mathbf{0}$, where

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) = f_0(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i f_i(\boldsymbol{x}) + \sum_{j=1}^{k} \mu_j h_j(\boldsymbol{x}).$$

- Since f_0, f_i's are convex, h_j's are affine, and $\lambda_i \geq 0$, by Theorem 5 in Lecture Note 14, $L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ is convex in $\boldsymbol{x}$. Combined with (4), $\boldsymbol{x}^*$ is a global minimum of $L(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ and

$$L\left(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*\right) \leq L\left(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*\right), \quad \forall \boldsymbol{x} \in \operatorname{dom} L.$$

  However, the left-hand side is just

$$L\left(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*\right) = f_0(\boldsymbol{x}^*) + \sum_{i=1}^{m} \underbrace{\lambda_i^* f_i(\boldsymbol{x}^*)}_{=0 \text{ by (3)}} + \sum_{j=1}^{k} \mu_j^* \underbrace{h_j(\boldsymbol{x}^*)}_{=0 \text{ by (1)}} = f_0(\boldsymbol{x}^*),$$

  therefore,

$$f_0(\boldsymbol{x}^*) \leq L\left(\boldsymbol{x}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*\right), \quad \forall \boldsymbol{x} \in \operatorname{dom} L.$$

- Set $\boldsymbol{x}$ to $\boldsymbol{y}$, we get

$$\begin{aligned} f_0(\boldsymbol{x}^*) &\leq L\left(\boldsymbol{y}, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*\right) \\ &= f_0(\boldsymbol{y}) + \sum_{i=1}^{m} \underbrace{\lambda_i^*}_{\geq 0 \text{ by (2)}} \cdot \underbrace{f_i(\boldsymbol{y})}_{\leq 0 \text{ by (1)}} + \sum_{j=1}^{k} \mu_j^* \cdot \underbrace{h_j(\boldsymbol{y})}_{=0 \text{ by feasibility}} \\ &\leq f_0(\boldsymbol{y}). \end{aligned}$$

  Since $\boldsymbol{y}$ is arbitrary, this proves that $\boldsymbol{x}^*$ is a global minimum.

- This proof didn't use Slater's condition. Slater's condition is needed in the necessity proof.

■

Consider arbitrary (convex or non-convex) optimization:

$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^n} \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \leq 0, \ i = 1, \ldots, m, \\ & h_j(\boldsymbol{x}) = 0, \ j = 1, \ldots, k. \end{aligned}$$

Here $f_0, \ldots, f_m$ could be non-convex, and $h_1, \ldots, h_k$ could be non-linear. We aim to convert this constrained optimization to unconstrained optimization by removing the constraints and softly penalizing them into the objective.

- Consider arbitrary penalty coefficients $\lambda_1, \ldots, \lambda_m$ and $\mu_1, \ldots, \mu_k$ with $\lambda_1, \ldots, \lambda_m \geq 0$:

$$\begin{aligned} \min \quad & f_0(\boldsymbol{x}), \\ \text{s.t.} \quad & f_i(\boldsymbol{x}) \leq 0, \ i = 1, \ldots, m, \\ & h_j(\boldsymbol{x}) = 0, \ j = 1, \ldots, k \end{aligned} \quad \rightarrow \quad \min \ f_0(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i f_i(\boldsymbol{x}) + \sum_{j=1}^{k} \mu_j h_j(\boldsymbol{x}).$$

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- Let's call the new objective *Lagrangian*:

$$L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \coloneqq f_0(\boldsymbol{x}) + \sum_{i=1}^{m} \lambda_i f_i(\boldsymbol{x}) + \sum_{j=1}^{k} \mu_j h_j(\boldsymbol{x}).$$

- If we minimize $L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ over $\boldsymbol{x}$, its solution may not satisfy the constraints $f_i(\boldsymbol{x}) \leq 0, h_j(\boldsymbol{x}) = 0$.
- However, its optimal objective value has a nice property.

Theorem 1. *Consider two optimization problems:*

$$\begin{aligned} \min_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f_0(\boldsymbol{x}), \\ s.t. \quad & f_i(\boldsymbol{x}) \leq 0, \ i = 1, \ldots, m, \\ & h_j(\boldsymbol{x}) = 0, \ j = 1, \ldots, k \end{aligned} \tag{1}$$

and

$$\min_{\boldsymbol{x}\in\mathbb{R}^n} \ L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}), \tag{2}$$

where $\boldsymbol{\lambda}, \boldsymbol{\mu}$ are arbitrary but $\boldsymbol{\lambda} \geq \mathbf{0}$. Then, the globally optimal objective value of the unconstrained optimization (2) is less than or equal to the globally optimal objective of the constrained optimization (1).

Proof. Let $\boldsymbol{x}^*$ denote a global minimum of (1) (the problem is non-convex but we don't care about its local minima) and $\tilde{\boldsymbol{x}}$ denote a global minimum of (2). We have

$$\begin{aligned} L(\tilde{\boldsymbol{x}}, \boldsymbol{\lambda}, \boldsymbol{\mu}) &= \min_{\boldsymbol{x}\in\mathbb{R}^n} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}) \\ &\leq L(\boldsymbol{x}^*, \boldsymbol{\lambda}, \boldsymbol{\mu}) \\ &= f_0(\boldsymbol{x}^*) + \sum_{i=1}^{m} \underbrace{\lambda_i}_{\geq 0} \underbrace{f_i(\boldsymbol{x}^*)}_{\leq 0} + \sum_{j=1}^{k} \mu_j \underbrace{h_j(\boldsymbol{x}^*)}_{=0} \\ &\leq f_0(\boldsymbol{x}^*), \end{aligned}$$

where the left-hand side is the optimal objective value of (2), and the right-hand side is the optimal objective value of (1). ■

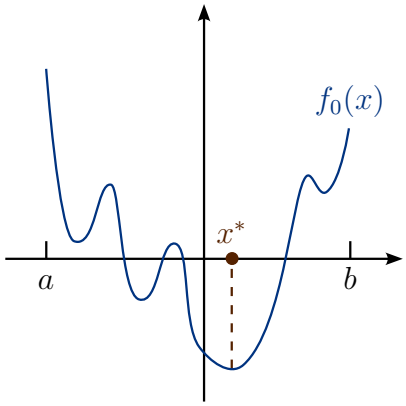
Example 2. Consider

$$\min_{x\in\mathbb{R}} \ f_0(x) \quad \text{s.t. } a \leq x \leq b.$$

We write this as

$$\begin{aligned} \min \quad & f_0(x), \\ \text{s.t.} \quad & a - x \leq 0, \\ & x - b \leq 0. \end{aligned}$$

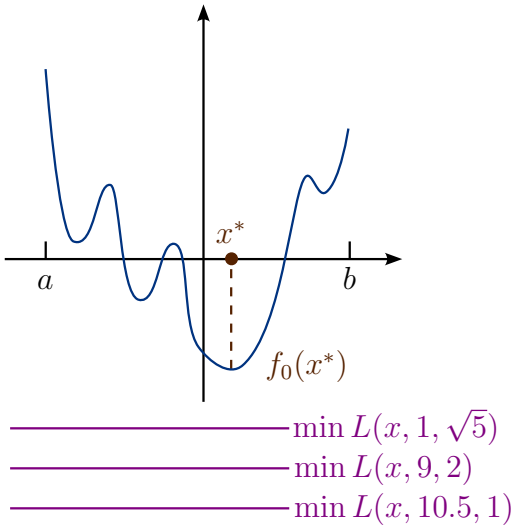
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We have

$$L(x, \lambda_1, \lambda_2) = f_0(x) + \lambda_1(a - x) + \lambda_2(x - b).$$

For each non-negative (λ_1, λ_2), we get a lower bound on $f_0(x^*)$:



Note: the solution $\tilde{x}$ to $\min L(x, \lambda_1, \lambda_2)$ may be greater than b or less than a (because we are not enforcing the constraints), but we only care about $L(\tilde{x}, \lambda_1, \lambda_2)$. ■

In Example 2, to find the tightest lower bound, we should optimize over our choices of λ_1, λ_2:

$$\max_{\lambda_1, \lambda_2 \geq 0} \text{lower bound} = \max_{\lambda_1, \lambda_2 \geq 0} \min_{x \in \mathbb{R}} L(x, \lambda_1, \lambda_2).$$

This makes us define the *dual function*:

$$\underbrace{\boxed{d(\boldsymbol{\lambda}, \boldsymbol{\mu})}}_{\text{dual function}} = \text{optimal objective value of (2)}$$

$$= \min_{\boldsymbol{x} \in \mathbb{R}^n} L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu}).$$

We optimize the lower bound of the optimal objective of (1) by maximizing $d(\boldsymbol{\lambda}, \boldsymbol{\mu})$ subject to $\boldsymbol{\lambda} \geq \mathbf{0}$.

Note: the domain of $d(\boldsymbol{\lambda}, \boldsymbol{\mu})$ is the intersection of domains of $f_0, f_1, \ldots, f_m, h_1, \ldots, h_k$ because $L(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{\mu})$ is a combination of all these functions and it should be well-defined.

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